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Prepare > Python > Numpy > Mean, Var, and Std

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Python

Mean, Var, and Std ★

Your Mean, Var, and Std submission got 20.00 points.

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- Problem
- Submissions
- Leaderboard
- Discussions
- Editorial

mean

The mean tool computes the arithmetic mean along the specified axis.

```
import numpy

my_array = numpy.array([ [1, 2], [3, 4] ])

print numpy.mean(my_array, axis = 0)      #Output : [ 2.  3.]
print numpy.mean(my_array, axis = 1)      #Output : [ 1.5  3.5]
print numpy.mean(my_array, axis = None)   #Output : 2.5
print numpy.mean(my_array)               #Output : 2.5
```

By default, the axis is None. Therefore, it computes the mean of the flattened array.

var

The var tool computes the arithmetic variance along the specified axis.

```
import numpy

my_array = numpy.array([ [1, 2], [3, 4] ])

print numpy.var(my_array, axis = 0)      #Output : [ 1.  1.]
print numpy.var(my_array, axis = 1)      #Output : [ 0.25  0.25]
print numpy.var(my_array, axis = None)   #Output : 1.25
print numpy.var(my_array)               #Output : 1.25
```

By default, the axis is None. Therefore, it computes the variance of the flattened array.

std

The std tool computes the arithmetic standard deviation along the specified axis.

```
import numpy

my_array = numpy.array([ [1, 2], [3, 4] ])

print numpy.std(my_array, axis = 0)      #Output : [ 1.  1.]
print numpy.std(my_array, axis = 1)      #Output : [ 0.5  0.5]
print numpy.std(my_array, axis = None)   #Output : 1.11803398875
print numpy.std(my_array)               #Output : 1.11803398875
```

By default, the axis is None. Therefore, it computes the standard deviation of the flattened array.

Author

[deleted]

Difficulty

Easy

Max Score

20

Submitted By

103668

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Task

You are given a 2-D array of size $N \times M$.

Your task is to find:

- The mean along axis 1

-
2. The var along axis **0**
3. The std along axis ***None***

Input Format

The first line contains the space separated values of ***N*** and ***M***.

The next ***N*** lines contains ***M*** space separated integers.

Output Format

First, print the mean.

Second, print the var.

Third, print the std.

Sample Input

```
2 2
1 2
3 4
```

Sample Output

```
[ 1.5  3.5]
[ 1.   1.]
1.11803398875
```

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Language

Pypy 3



```
1
2
3
4
5
6
7
8
9
10
```